# 8 DYNAMICS AS RELATION

*Sooner or later*
*the exact measurement of our time*
*will resume—*
*and we'll be on the ship that's bound*
*for the bitterest shore. (II, 9)*

How does one describe a world in which everything occurs but there is no time variable? In which there is no common time and no privileged direction in which change occurs?

In the simplest way, the same way that we had thought about the world until Newton convinced us all that a variable time was indispensable.

To describe the world, the time variable is not required. What is required are variables that actually describe it: quantities that we can perceive, observe, and eventually measure. The length of a road, the height of a tree, the temperature of a forehead, the weight of a piece of bread, the color of the sky, the number of stars in the celestial vault, the elasticity of a piece of bamboo, the speed of a train, the pressure of a hand on a shoulder, the pain of a loss, the position of the hands on a clock, the height of the sun in the sky . . . These are the terms in which we describe the world. Quantities and properties that we see continuously *changing.* In these changes there are regularities: a stone falls faster than a feather. Sun and moon circle after each other in the sky, passing by each other once a month. . . . Among these quantities there are some that we see changing regularly with respect to others: the number of days, the phases of the moon, the height of the sun on the horizon, the position of the hands of a clock. It is useful to employ *these* as points of reference: let's meet three days after the next full moon, when the sun is at its highest in the sky. I'll see you tomorrow, when the clock shows 4:35. If we find a sufficient number of variables that remain synchronized enough in relation to each other, it is convenient to use them in order to speak of *when*.

There is no need in any of this to choose a privileged variable and call it "time." What we need, if we want to do science, is a theory that tells us how the variables change with respect to each other. That is to say, how one changes when others change. The fundamental theory of the world must be